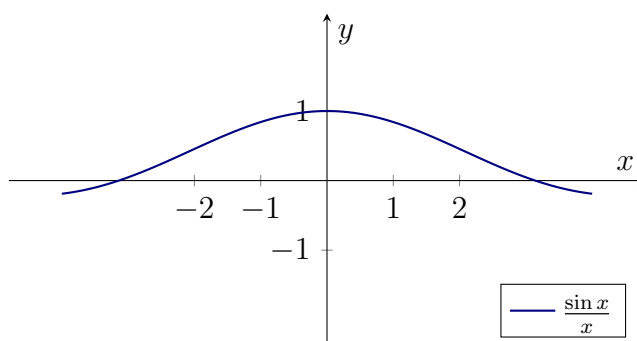


Topic 4: Limits¹

LIMIT OF A FUNCTION

Calculus, developed in the 1670s, involves the use of limits of functions. However, the definition of limits was not initially rigorous. The modern concept of a function's limit traces back to Bernard Bolzano's epsilon-delta definition of limits in the 1820s. To see this, consider the following example:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$



The $\frac{\sin x}{x}$ is undefined when $x = 0$. However, we kind of know that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ by looking at the graph. The question for us today is: How can we *prove* $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$?

The proof is based on the formal definition of limits. An intuitive definition of $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ from freshmen calculus is:

As x gets close to 0, $\frac{\sin x}{x}$ becomes very close to 1.

However, this is not rigorous enough for a proof. What is the rigorous definition of “getting close to”? Bolzano, Cauchy, and Weierstrass established the modern definition of limits:

$\frac{\sin x}{x}$ can get **arbitrarily** close to 1, as long as x is close **enough** to 0.

¹Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.

Definition 1 (Epsilon-Delta Definition of a Limit)

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a , and let L be a real number.

We say that

$$\lim_{x \rightarrow a} f(x) = L$$

if, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta$$

If $\lim_{x \rightarrow a} f(x) = L$, we say that $f(x)$ **converges** to L as x approaches a .

We need to show that for any given ε , there exists a $\delta > 0$ which satisfies the condition in Definition 1. When developing proofs using the epsilon-delta definition, keep in mind that δ can change with ε . Typically, the proof involves expressing $\delta(\varepsilon)$ as a function of ε that satisfies the epsilon-delta definition.

Example 1

Use the epsilon-delta definition of a limit to prove that

$$\lim_{x \rightarrow 2} x^2 = 4$$

In this example, $L = 4$, $a = 2$, and $f(x) = x^2$.

Step 1: Choose an open interval containing a .

Let's consider $x \in (1, 3)$.

Step 2: Start with the inequality $|f(x) - L| < \varepsilon$.

$$\begin{aligned} |f(x) - L| &= |x^2 - 4| = |(x + 2)||x - 2| \\ &= (x + 2)|x - 2| \\ &< 5|x - 2| \end{aligned}$$

The first line is by algebra. The second line holds because we consider $x \in (1, 3)$. The third line holds because we consider $x \in (1, 3)$.

Now, suppose we are given an arbitrary $\varepsilon > 0$. How can we guarantee that $|f(x) - L| < \varepsilon$ holds?

The conditions we need are:

$$5|x - 2| < \varepsilon \quad \text{and} \quad x \in (1, 3)$$

We need the second part $x \in (1, 3)$ because the result of $|f(x) - L| < 5|x - 2|$ relies on $x \in (1, 3)$ (it does not hold for any $x \in \mathbb{R}$).

Step 3: Choose $\delta > 0$ to make the inequality hold.

The inequality $|f(x) - L| < \varepsilon$ holds if

$$5|x - 2| < \varepsilon \quad \text{and} \quad x \in (1, 3)$$

which is equivalent to

$$|x - 2| < \frac{\varepsilon}{5} \quad \text{and} \quad x \in (1, 3)$$

which is equivalent to

$$|x - 2| < \frac{\varepsilon}{5} \quad \text{and} \quad |x - 2| < 1$$

Therefore, we can choose $\delta = \min\{1, \frac{\varepsilon}{5}\}$. The inequality $|f(x) - L| < \varepsilon$ holds if $\delta = \min\{1, \frac{\varepsilon}{5}\}$.

When we are given an **arbitrary** $\varepsilon > 0$, no matter how small ε is, we can always choose a $\delta = \min\{1, \frac{\varepsilon}{5}\}$ **accordingly** to have:

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta$$

□

Example 2

Use the epsilon-delta definition of a limit to prove that

$$\lim_{x \rightarrow 1} \frac{x}{x^2 + 1} = \frac{1}{2}$$

ONE-SIDED LIMITS**Definition 2** (Right-sided limit)

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a , and let R be a real number.

We say that

$$\lim_{x \rightarrow a^+} f(x) = R$$

if, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - R| < \varepsilon \quad \text{whenever} \quad 0 < x - a < \delta$$

Definition 3 (Left-sided limit)

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a , and let L be a real number.

We say that

$$\lim_{x \rightarrow a^-} f(x) = L$$

if, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < a - x < \delta$$

Example 3

Find the right-sided limit and the left-sided limit of

$$\frac{x^2 - 1}{|x - 1|}$$

as x approaches 1.

Let us consider the behavior of

$$f(x) = \frac{x^2 - 1}{|x - 1|} = \frac{(x - 1)(x + 1)}{|x - 1|}$$

separately for x approaching 1 from the right ($x \rightarrow 1^+$) and from the left ($x \rightarrow 1^-$).

• **Right-sided limit:**

When $x \rightarrow 1^+$, $x > 1$, so $|x - 1| = x - 1$. The expression becomes:

$$\frac{(x-1)(x+1)}{|x-1|} = \frac{(x-1)(x+1)}{x-1} = x+1$$

As $x \rightarrow 1^+$, $x+1 \rightarrow 2$. Therefore, the right-sided limit is:

$$\lim_{x \rightarrow 1^+} \frac{x^2 - 1}{|x - 1|} = 2$$

• **Left-sided limit:**

When $x \rightarrow 1^-$, $x < 1$, so $|x - 1| = -(x - 1)$. The expression becomes:

$$\frac{(x-1)(x+1)}{|x-1|} = \frac{(x-1)(x+1)}{-(x-1)} = -(x+1)$$

As $x \rightarrow 1^-$, $-(x+1) \rightarrow -2$. Therefore, the left-sided limit is:

$$\lim_{x \rightarrow 1^-} \frac{x^2 - 1}{|x - 1|} = -2$$

LIMITS AT INFINITY

Definition 4

Let $f(x)$ be defined on an open interval in $\mathbb{R}$, and let L be a real number.

(1) We say that

$$\lim_{x \rightarrow \infty} f(x) = L$$

if, $\forall \varepsilon > 0$, $\exists c > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad x > c$$

(2) We say that

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if, $\forall \varepsilon > 0$, $\exists c > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad x < -c$$

Example 4

Use the epsilon-delta definition of a limit to prove that

$$\lim_{x \rightarrow -\infty} e^x = 0$$

(Hint: $e^x < \varepsilon$ is equivalent to $x < \ln(\varepsilon)$ for all $\varepsilon > 0$.)

EXISTENCE OF LIMIT

The epsilon-delta definition of a limit highlights the need for $f(x)$ to get arbitrarily close to a single value L as x approaches a from both sides.

There are three cases where the limit of $f(x)$ **does not exist**.

- **Case 1** Different Left-sided and Right-sided Limits

Example: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{|x - 1|}$

- **Case 2** Unbounded Behavior (Infinity)

Example: $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$.

- **Case 3** Oscillatory Behavior

Example: $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$.

Example 5

Evaluate the limits at $x = 0$ for the following functions f , if they exist.

1. $f(x) = |x|$

2. $f(x) = x^2$

3. $f(x) = \frac{1}{x^2}$

4. $f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$

SOME LIMIT THEOREMS

The Squeeze Theorem (also known as the Sandwich Theorem) is a method to find the limit of a function that is “squeezed” between two other functions with known and equal limits.

Theorem 1: Squeeze Theorem

Let $f(x), g(x), h(x)$ be defined on an open interval containing a , and let L be a real number. Suppose $f(x) \leq g(x) \leq h(x)$.

If

$$\lim_{x \rightarrow a} f(x) = L \text{ and } \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

The Squeeze Theorem is useful when $g(x)$ is difficult to analyze directly, but is squeezed between two simpler functions, $f(x)$ and $h(x)$, whose limits are easier to determine.

Example 6

Use the Squeeze Theorem to prove that

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^4 + 1} = 0$$

In this example, $g(x) = \frac{x^2}{x^4 + 1}$.

Step 1: Choose $f(x)$ and $h(x)$ to “squeeze” $g(x)$.

We want some $f(x)$ and $h(x)$ that satisfy $f(x) \leq g(x) \leq h(x)$. More importantly, the limits of $f(x)$ and $h(x)$ as $x \rightarrow a$ are **simple** and **identical** (both are L).

Let $f(x) = 0$. Clearly, $f(x) \leq g(x)$ for all $x > 0$, and $\lim_{x \rightarrow \infty} f(x) = 0$.

Let $h(x) = \frac{x^2}{x^4} = \frac{1}{x^2}$ for all $x > 0$. Then, $g(x) \leq h(x)$ and $\lim_{x \rightarrow \infty} h(x) = 0$.

Step 2: Apply the Squeeze Theorem.

By the Squeeze Theorem, $\lim_{x \rightarrow \infty} \frac{x^2}{x^4 + 1} = 0$. $\square$

Example 7

Use the Squeeze Theorem, prove that

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x = 1$$

(Hint: $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$ and $\lim_{x \rightarrow \infty} e^{\frac{1}{x}} = 1$)

The following theorems are straightforward. We will use them frequently in econometrics.

Theorem 2

Let $f(x), g(x)$ be defined on an open interval containing a , and let L_f, L_g be real numbers.

Suppose

$$\lim_{x \rightarrow a} f(x) = L_f \text{ and } \lim_{x \rightarrow a} g(x) = L_g$$

Then, we have:

$$\lim_{x \rightarrow a} [f(x) + g(x)] = L_f + L_g$$

$$\lim_{x \rightarrow a} [f(x)g(x)] = L_f L_g$$

$$\lim_{x \rightarrow a} [f(x)/g(x)] = L_f/L_g, \quad L_g \neq 0$$

LIMIT OF A SEQUENCE

We have discussed the limit of a function. Sometimes we will also work on sequences, and there is a similar epsilon-delta definition of the limit of a sequence. The limit of a sequence is frequently used in deriving a game-theoretic equilibrium.²

We write a **sequence** as $\{a_n\}$, where a_n represents the n -th term of the sequence. For example, the sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$ can be written as $\{a_n\}$ where $a_n = \frac{1}{n}$.

²We will study the Perfect Bayesian Equilibrium and Sequential Equilibrium in the core courses.

Definition 5 (Limit of a Sequence)

Let $\{a_n\}$ be a sequence, and let L be a real number. We say that

$$\lim_{n \rightarrow \infty} a_n = L$$

if, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that

$$|a_n - L| < \varepsilon \quad \text{whenever } n \geq N$$

If $\lim_{n \rightarrow \infty} a_n = L$, we say that the sequence $\{a_n\}$ is **convergent**, or it **converges** to the limit L .

The squeeze theorem and the theorems in Theorem 2 also apply for the limit of a sequence.

Example 8

Use the Definition above to prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Step 1: Start with the inequality $|a_n - L| < \varepsilon$.

$$|a_n - L| = \left| \frac{1}{n} \right| = \frac{1}{n}$$

The second equality holds because we consider $n \in \mathbb{N}$.

Now, suppose we are given an arbitrary $\varepsilon > 0$. How can we guarantee that $|a_n - L| < \varepsilon$ holds? The conditions we need is:

$$\frac{1}{n} < \varepsilon$$

Step 2: Choose $N \in \mathbb{N}$ to make the inequality hold.

The inequality $|a_n - L| < \varepsilon$ holds if

$$\frac{1}{n} < \varepsilon$$

which is equivalent to

$$n > \frac{1}{\varepsilon}$$

Note that we are choosing $N \in \mathbb{N}$ (but not $N \in \mathbb{R}$) to satisfy the definition of limit of a sequence. Therefore, we cannot just choose $N = \varepsilon$ because ε might not be an integer.

Therefore, we can choose

$$N = \left\lceil \frac{1}{\varepsilon} \right\rceil$$

where $\lceil \cdot \rceil$ is the ceiling function to round up to the nearest integer.

When we are given an **arbitrary** $\varepsilon > 0$, no matter how small ε is, we can always choose a $N = \left\lceil \frac{1}{\varepsilon} \right\rceil$ **accordingly** to have:

$$|a_n - L| < \varepsilon \quad \text{whenever } n \geq N$$

□

CAUCHY'S CRITERION

Notice that in the example of proving $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, we need to know the limit is 0 to begin with our proof. More generally, we need to know the number L in $\lim_{n \rightarrow \infty} a_n = L$. Is there a way to prove that $\{a_n\}$ is convergent without knowing L ?

Definition 6 (Cauchy's Criterion)

Let $\{a_n\}$ be a sequence.

We say that $\{a_n\}$ satisfies Cauchy's Criterion (or $\{a_n\}$ is a Cauchy sequence) if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon \quad \text{whenever } n, m \geq N$$

Theorem 3

A sequence is convergent *if and only if* it satisfies Cauchy's criterion.^a

^aIn $\mathbb{R}^n$ space.

Example 9

Use Cauchy's criterion to prove that the sequence $\{a_n\}$ where $a_n = \frac{1}{n}$ is convergent.

Step 1: Start with the inequality $|a_n - a_m| < \varepsilon$.

Let n and m be any natural numbers greater than N . Without loss of generality (WLOG), let $n \leq m$.³ We have:

$$\begin{aligned} |a_n - a_m| &= \left| \frac{1}{n} - \frac{1}{m} \right| \\ &= \frac{1}{n} - \frac{1}{m} \\ &< \frac{1}{n} \end{aligned}$$

Step 2: Choose $N \in \mathbb{N}$ to make the inequality hold.

The inequality $|a_n - a_m| < \varepsilon$ holds if

$$\frac{1}{n} < \varepsilon$$

which is equivalent to

$$n > \frac{1}{\varepsilon}$$

Therefore, we can choose

$$N = \left\lceil \frac{1}{\varepsilon} \right\rceil$$

where $\lceil \cdot \rceil$ is the ceiling function to round up to the nearest integer.

Compare this proof with Exercise 8. The difference here is that we don't need to know which L is the sequence $\{a_n\}$ converging to. $\square$

Example 10

Use Cauchy's criterion to prove that the sequence $\{a_n\}$ where $a_n = \frac{1}{n^2}$ is convergent.

³“WLOG” is commonly used in proofs. Suppose it is $m \leq n$ instead of $n \leq m$, we can just write m as n and write n as m , so that $n \leq m$ holds. This is why we do not lose generality.

REFERENCES

- Apostol, T. (1957). *Mathematical Analysis*. Addison-Wesley series in mathematics. Addison-Wesley.
- Rudin, W. (1964). *Principles of Mathematical Analysis*. International series in pure and applied mathematics. McGraw-Hill.